

Dynamic Volatility Weighting: Economic Value and Controls

Eddie Liu | Separate legacy stock-model experiment

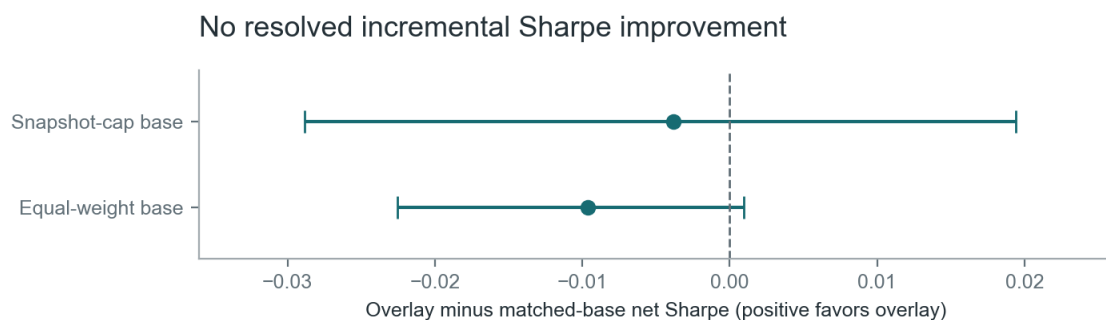
Research finding

The monthly overlay has net Sharpe **1.115** versus **1.119** for the same snapshot-cap base without the tilt. The difference is -0.0038, with a paired 95% temporal interval [-0.0289, +0.0194]. This diagnostic does not establish incremental Sharpe improvement.

Important: later surviving constituents and a later cap snapshot are used throughout history. These are conditional accounting results, not point-in-time alpha or achievable-return estimates. Cached forecast training was not rerun.

Monthly, 10 bps	Net CAGR	Net Sharpe	Max DD
Snapshot-cap base	26.02%	1.119	-33.25%
Snapshot-cap + overlay	25.95%	1.115	-32.51%
Equal-weight base	15.54%	0.738	-38.34%
Equal-weight + overlay	15.33%	0.728	-38.18%

Table 1. 2,390 common return dates, 2015-07-06 to 2024-12-31. Next-close execution, monthly rebalance, 10 bps per dollar bought/sold. CAGR is geometric; Sharpe subtracts a constant assumed daily risk-free return of 0.0001. Final positions are not liquidated.



Paired stationary-bootstrap intervals: mean block 21, 2,000 resamples. Both include zero. They condition on the cached forecasts and biased sample and do not repair source bias.

What is being tested

The legacy stock model multiplies GARCH volatility by an LSTM ratio correction. Weights are proportional to base weights times clipped trailing-volatility / saved-forecast ratios (0.5-1.5). It is not the index additive-residual LSTM in Part A; the two studies must not be interpreted as one prediction-to-P&L experiment.

Portfolio Mechanics and Sensitivities

Accounting and information timing

Signals formed after close d execute at close $d+1$, earning the return ending $d+2$. Weights drift between trades. The engine compounds weighted simple stock returns, charges actual buy/sell notional, and solves post-cost allocations self-consistently. Rebalances use actual observed trading sessions, not calendar-day filters. Missing held-asset returns fail rather than silently become zero.

Net Sharpe by rebalance frequency

10 bps, one-close lag	Matched cap base	Cap overlay	Difference
Daily	1.111	0.937	-0.1746
Weekly	1.122	1.083	-0.0392
Monthly	1.119	1.115	-0.0038
Yearly	1.180	1.182	+0.0024

Monthly was fixed as primary. The small positive yearly point estimate is not selected as the headline outcome; frequency sensitivity is not a fresh holdout test.

Net Sharpe by transaction-cost assumption

Monthly cost / traded \$	Matched cap base	Cap overlay	Difference
0 bps	1.122	1.126	+0.0038
10 bps	1.119	1.115	-0.0038
25 bps	1.114	1.098	-0.0154

Annualized traded notional is 0.73x wealth for the base and 2.29x for the overlay. The small gross Sharpe advantage disappears under the primary cost assumption. The idealized same-close sensitivity also does not improve monthly net Sharpe.

Data quality, scope, and reproducibility

The panel has 503 stock columns and one repeated cap-weight row. 4,616 eligible asset-days receive neutral tilts, including 76 nonpositive forecasts. Historical membership, delistings, contemporaneous caps, and independent per-fit forecast records remain unavailable. Equal weighting does not remove survivor bias.

The ^GSPC price index is context only: it is not a matched total-return comparator for adjusted-close stock portfolios. A same-universe GARCH-only panel is also missing, so these comparisons cannot isolate an LSTM contribution. No execution-capacity test is claimed.

Code: `src/evaluation/portfolio.py`; runner: `scripts/evaluate_portfolio_overlay.py`; tests: `tests/test_portfolio.py`; protocol: `docs/portfolio-overlay-protocol.md`. `reports/portfolio_overlay_results.json` stores scores, sensitivities, hashes and monthly portfolio aggregates. Raw inputs and daily position-level data are not distributed.

Based on a collaborative research project; portfolio report prepared by Eddie Liu.